Tennessee Valley Authority, Sequoyah Nuclear Plant, P.O. Box 2000, Soddy Daisy, Tennessee 37384

July 5, 2016

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Sequoyah Nuclear Plant, Units 1 and 2
Renewed Facility Operating License Nos. DPR-77 and DPR-79
NRC Docket Nos. 50-327 and 50-328

Subject: **Licensee Event Report 50-327 and 50-328/2016-003-00, Control Room Door Unable to Close Causes Inoperable Control Room Envelope**

The enclosed Licensee Event Report provides details concerning the inability to close a main control room door resulting in both trains of the Control Room Emergency Ventilation System being declared inoperable. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact Michael McBrearty, Site Licensing Manager, at (423) 843-7170.

Respectfully,

For Christopher J. Schwarz
Site Vice President
Sequoyah Nuclear Plant

Enclosure: Licensee Event Report 50-327 and 50-328/2016-003-00
cc: NRC Regional Administrator – Region II
NRC Senior Resident Inspector – Sequoyah Nuclear Plant

NRC FORM 366 (11-2015)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018

LICENSEE EVENT REPORT (LER)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Sequoyah Nuclear Plant Unit 1	05000327	1 OF 6

4. TITLE
Control Room Door Unable to Close Causes Inoperable Control Room Envelope Boundary

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: Sequoyah Nuclear Plant Unit 2	DOCKET NUMBER: 05000328
05	03	2016	2016 -	003 -	00	07	05	2016	FACILITY NAME: NA	DOCKET NUMBER:

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
100	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ **50.73(a)(2)(v)(D)**	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Scott T Bowman	423-843-6910

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	NA	DR	PRECISION HARDWARE, INC	N					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On May 3, 2016, at 0833 Eastern Daylight Time, Security received an open door alarm regarding a Main Control Room (MCR) door. At 0847, Security notified the MCR staff of the door alarm and that the door was incapable of closure. At this time, both Control Room Emergency Ventilation System (CREVS) trains were declared inoperable in accordance with Technical Specification (TS) 3.7.10, Condition B, due to the inoperability of the Control Room Envelope (CRE). Attempts to close the door were made, and it was identified that a screw had become wedged at the base of the door preventing it from latching. At 0855, the screw was removed and the door verified to close as designed. CREVS was determined to be operable and TS 3.7.10, Condition B was exited.

The screw was the bottom screw from the door latch guard and appears to have stripped loose and fallen into the rubber seal around the base of the MCR door frame preventing closure of the door. Corrective actions include replacing the missing screw on the door latch guard and evaluating existing preventive maintenance instructions associated with CREVS boundary doors to ensure screws are tightened, as needed, on a routine basis.

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LICENSEE EVENT REPORT (LER) CONTINUATION SHEET		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.	

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Sequoyah Nuclear Plant Unit 1	05000327	2016	- 003 -	00

NARRATIVE

I. PLANT OPERATING CONDITIONS BEFORE THE EVENT

At the time of the event, Sequoyah Nuclear Plant (SQN) Unit 1 and Unit 2 were in Mode 1 at 100 percent rated thermal power.

II. DESCRIPTION OF EVENTS

A. Event:

On May 3, 2016, at 0833 Eastern Daylight Time (EDT), Security received an open door [EIIS Code DR] alarm [EIIS Code ALM] on a Main Control Room [EIIS Code NA] (MCR) door. At 0847, Security notified the MCR staff that the door would not maintain its seal and would not latch. Security remained in the vicinity of the door the remainder of the time it was unable to close. The leak was approximately a one-eighth-inch opening across the entire sealing surface [SEAL] of the door. As a result, both units entered Technical Specification (TS) 3.7.10, Control Room Emergency Ventilation System [EIIS Code VI](CREVS), Condition B, due to the inoperability of the Control Room Envelope (CRE).

During inspection of the sealing surface, a screw was found wedged under the rubber seal on the bottom edge of the door. The screw was the bottom screw from the door latch guard. At 0855, the screw was removed and the door was verified to close as designed. CREVS was declared operable and TS 3.7.10, Condition B, was exited.

At 1513, an 8-hour non-emergency event notification (EN 51900) was made to the NRC in accordance with 10 CFR 50.72(b)(3)(v)(D), as an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident. This LER is submitted based on NUREG-1022, Revision 3, Section 3.2.7 guidance which identifies that the requirements of 10 CFR 50.73(a)(2)(v)(D) apply when a system that is used to mitigate the consequences of an accident was declared TS inoperable and no redundant system or equipment could be declared operable. Engineering evaluation determined the requirement to support long term occupancy would not have been affected to the point that would have resulted in exceeding the design basis dose assumptions for the MCR operators. Therefore, no loss of safety function occurred.

The screw was the bottom screw from the door latch guard and appears to have stripped loose and fallen out.

B. Status of structures, components, or systems that were inoperable at the start of the event and contributed to the event:

No inoperable structures, components, or systems contributed to this event.

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C. Dates and approximate times of occurrences:

Date/Time (EDT)	Description
05/03/16, 0833	Security received an open door alarm on a MCR door.
05/03/16, 0847	Security notified MCR staff of the door alarm and the door was incapable of closure. Both units enter TS 3.7.10, Condition B, due to the inoperability of the CRE.
05/03/16, 0855	After investigation, a screw was found, removed, and the door verified to close as designed. Both units exit TS 3.7.10, Condition B.

D. Manufacturer and model number of each component that failed during the event:

The failed component was associated with the door latch guard for the MCR door. The door latch guard is model number 1625, manufactured by Precision Hardware, Inc.

E. Other systems or secondary functions affected:

There were no systems or secondary functions affected by this event.

F. Method of discovery of each component or system failure or procedural error:

Security received an open door alarm associated with the MCR door.

G. The failure mode, mechanism, and effect of each failed component, if known:

The screw that prevented the MCR door from closing was the bottom screw from the door latch guard and appears to have stripped loose and fallen into the rubber seal around the base of the MCR door frame preventing the door from closing.

H. Operator actions:

Following notification of the issue to the MCR operators, both CREVS trains were declared inoperable in accordance with TS 3.7.10, Condition B, due to the inoperability of the CRE. MCR staff contacted individuals to evaluate and repair as necessary. Additionally, mitigating actions were initiated and met by verifying the availability of Potassium Iodine and self contained breathing apparatus to MCR operators within the CRE.

I. Automatically and manually initiated safety system responses:

There were no automatic or manual system responses associated with this event.

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III. CAUSE OF THE EVENT

A. The cause of each component or system failure or personnel error, if known:

The cause of the event was the bottom screw in the door latch guard appears to have stripped loose and fallen into the rubber seal around the base of the MCR door frame preventing closure of the door.

B. The cause(s) and circumstances for each human performance related root cause:

There was no human performance related root cause.

IV. ANALYSIS OF THE EVENT

The CREVS is a safety related Engineered Safety Features system required to operate during design basis accidents (DBAs) to mitigate the radiological consequences to the control room operators. The CREVS is designed to maintain a habitable environment in the CRE for a mission time of 30 days of continuous occupancy after a DBA without exceeding a 5 rem whole body dose or its equivalent to any part of the body. The operability of the CRE boundary must be maintained to ensure that the inleakage of unfiltered air into the CRE will not exceed the inleakage assumed in the licensing basis analysis of DBA consequences to CRE occupants. Additionally, the CRE boundary ensures occupants are protected from hazardous chemicals and smoke.

Engineering evaluation concluded that the requirement to support long term occupancy would not have been affected to the point that would have resulted in exceeding the design basis dose assumptions for the control room operators as the assumed unfiltered air inleakage would not have been exceeded. Therefore, the CREVS and the CRE would have been able to fulfill the required 30 days of continuous occupancy after a DBA.

V. ASSESSMENT OF SAFETY CONSEQUENCES

The temporary loss of ability to close the MCR door did not challenge nuclear or radiological safety. No actual loss of safety function occurred. The CREVS maintained the ability to limit MCR operator doses below the required limits.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event:

Although the MCR door was ajar, the CREVS maintained the ability to limit MCR operator doses below the required limits.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident:

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The event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from discovery of the failure until the train was returned to service:

The elapsed time from discovery of the open door rendering the CRE inoperable until the CRE was restored to operable status was 22 minutes.

VI. CORRECTIVE ACTIONS

This event was entered into the Tennessee Valley Authority Corrective Action Program under Condition Report (CR) 1166927.

A. Immediate Corrective Actions:

MCR Operators contacted various organizations to investigate why the door would not close. During inspection of the sealing surface around the MCR door, a screw was found and removed. The MCR door was closed and TS 3.7.10, Condition B, was exited for both units.

B. Corrective actions to reduce probability of similar events occurring in the future:

Corrective actions include evaluating existing preventive maintenance instructions associated with CREVS boundary doors to ensure screws are tightened, as needed, on a routine basis.

Additional actions include:

Initiating a work order to replace the missing screw on the door latch guard.

VII. ADDITIONAL INFORMATION

A. Previous similar events at the same plant:

There have been no similar events at SQN in the past three years with the same underlying cause.

B. Additional Information:

None.

C. Safety System Functional Failure Consideration:

This condition did not result in a safety system functional failure as defined in NEI 99-02.

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D. Scrams with Complications Consideration:

There was no scram associated with this event.

VIII. COMMITMENTS

None.